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# Learning-based Multifunctional Elbow Exoskeleton Control

Xiaofeng Xiong, Cao Danh Do, and Poramate Manoonpong

***Abstract*—We propose a learning-based model for multifunctional elbow exoskeleton control, i.e., assist- and resist-as-needed (AAN and RAN). The model consists of online iterative learning and impedance adaptation mechanisms for predictive and variable compliant joint control. The model was implemented on a lightweight (0.425 $kg$) and portable elbow exoskeleton (i.e., POW-EXO) worn by three subjects, respectively. The implementation relies only on internal pose (e.g., joint position) feedback, rather than physical compliant mechanisms (e.g., springs) and external sensors (e.g., electromyography (EMG) or force) typically required by conventional exoskeletons and controllers. The proposed model provides a novel technique to achieve multifunctional exoskeleton control with minimal mechatronics and sensing. Interestingly, its RAN control and POW-EXO as a quantification means, may reveal interactive (mechanical) impedance variance and invariance in human motor control.**

***Index Terms*—Robotics and mechatronics, wearable robots, force control, variable compliant control.**

## I. Introduction

ELBOW assistance and exercise are important for restoring patients' motor functions and preventing athletes' injuries [1], [2]. Mechatronic systems (e.g., exoskeletons) have proven to be an effective means for improving performance and reducing economic costs [3], [4]. Exoskeletons can provide human limbs not only with assistant joint torques in rehabilitation, but also resistant torques for exercise. Although there are exoskeleton products and prototypes on markets and in labs [5], there is no learning-based model for portable multifunctional elbow exoskeleton control that online adapts to different wearers and tasks.

Most existing models focus on a single function, i.e., assistance or resistance [6], [7]. For instance, many models (e.g., reinforcement learning) have been developed to learn assistant strategies for upper limb exoskeleton control [8], [9]. However, none of these address the issue of portable multifunctional exoskeleton control with online impedance adaptation, i.e., assist- and resist-as-needed (i.e., AAN and RAN) functions. The functions have been implemented to provide compliant interactions between exoskeletons and wearers based on impedance and admittance control [10]. Impedance control produces exoskeleton joint torques based on pose (e.g., position) feedback, while admittance control generates joint positions in terms of force feedback [11].

Adaptive impedance/admittance is more suitable for exoskeleton control [12], compared to a fixed one for a single application. This is because such an adaptation can aid various wearers in different tasks (i.e., assistance and resistance). Additionally, in contrast to traditional controllers [10], [13], the adaptation enables exoskeletons to reach a compromise between tasks (e.g., trajectory tracking) and compliance. However, most existing learning impedance models rely heavily on additional external sensing such as EMG and force [14], [15]. Force instance, Meattini et al. (2020) developed an EMG-driven method to close a human-robot control loop [16], thereby preventing complex pre-training procedures and imprecise force prediction for various wearers [17]. However, their method is subject to intrinsic sensing limitations such as prediction inaccuracy and abnormal EMG-torque relationship (see details in section IV). Although some limitations can be addressed by introducing novel sensors, e.g., for measuring muscle circumference [12], their design and calibration are sophisticated. Such sophistication reduces usability and practicality in different applications. A detailed review of related compliant control and designs is presented in [5], [10], [18]. Taken together, there is no advanced learning model for predictive compliant multifunctional elbow exoskeleton control without external sensing (e.g., EMG) and passive mechanisms (e.g., springs).

To address the problem, we propose a learning-based model to free compliant multifunctional exoskeleton control from external sensing (e.g., EMG) and passive compliant mechanisms (e.g., springs). The model consists of iterative learning and online impedance mechanisms for predictive and variable compliant exoskeleton control. The implementation relies only on internal position and velocity feedback. The proposed model was validated on three healthy subjects wearing an elbow exoskeleton. 'Multifunctional' here means that an exoskeleton can provide not only assist-as-needed (AAN) physical support, but also resist-as-needed (RAN) exercises; while typical exoskeleton control has been developed for a specific function, e.g., assistance [17]. The proposed AAN here

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focuses on assistant adaptations to various wearers, thereby online compensating for wearers' involuntary (i.e., reflex) involvement. This is different from traditional control which aims at maximizing wearers' involvement [13]. In contrast to the AAN, the proposed RAN here aims to maximize wearer's involvement in exercises. Note that the presented control and lightweight hardware are developed for portable assistance and resistance, rather than a single function performed by heavy multi-joint ground-based exoskeletons [18]. To the best of our knowledge, the proposed model is the first to achieve learning-based multifunctional control of a portable exoskeleton with minimal (i.e., internal) sensing. The proposed model contributes to state-of-art (SOA) by providing intelligent motion control for exoskeleton assistant and resistant tasks with minimal sensory feedback. It is a novel supplement to SOA exoskeleton designs and control [15], [19].

The remainder of the paper is structured as follows. The proposed learning-based control model is presented in section II. Section III contains the experimental results of assistant and resistant tasks. A discussion and conclusion is provided in section IV.

## II. Learning-based Model

The model consists of iterative learning and online impedance mechanisms for predictive and adaptive exoskeleton control. The model was validated using a lightweight (0.425 $kg$) wearable elbow exoskeleton (i.e., POW-EXO). The POW-EXO consists of 3D printed polylactic acid (PLA) parts, aluminum connectors, and a Dynamixel actuator (see Fig. 1 and Table I). Detail of the exoskeleton design can be seen in [20].

TABLE I
POW-EXO SPECIFICATION

| Weight | Actuator | Stall torque | Voltage |
|---|---|---|---|
| 0.425 $[kg]$ | XM430 | 4.1 $[N \cdot m]$ | 12.0 $[V]$ |

The inverse dynamics model of the POW-EXO and its wearer (see Fig. 1) is given by [17],

$$I\ddot{q} + h(\dot{q}) + g(q) = \tau + \tau_h = \tau_s, \tag{1}$$

where $I$, $h$, and $g$ represent the unknown inertial, viscous, and gravity terms determined by exoskeleton joint acceleration $\ddot{q}$, velocity $\dot{q}$, and position $q$. The control input $\tau$ is generated by the learning-based model (see Fig. 2), compensating for the unknown exoskeleton dynamics (e.g., $g$) and interaction torque $\tau_h$. $\tau$ is given by,

$$\tau = -(\tau_{ff} + \tau_{fb}), \tag{2}$$

where $\tau_{ff}$ is iteratively learned for the intrinsic exoskeleton dynamics, while $\tau_{fb}$ is online adapted to the unknown interaction torque $\tau_h$. Minus (i.e., '$-$') indicates the assistant torque $\tau$ (i.e., anti-clockwise, see the green arrow in Fig. 1).

### A. Iterative Learning Mechanism (ILM)

Iterative learning is a well-studied research field in robot-assisted applications. The ILM follows the proverb "practice makes perfect," facilitating better task performance than pure feedback control. This is because the ILM learns from previous trials to improve task performance. Moreover, it does not require much knowledge of controlled dynamics to design its learning gain. This advantage endows the ILM with greater flexibility in exoskeleton control, compared to feedback control. This is because the dynamics of an exoskeleton and its wearer intrinsically vary in accordance with different tasks. Therefore, unknown exoskeleton dynamics are compensated by iteratively modulating the feedforward term $\tau_{ff}$ of the control input $\tau$,

$$\tau_{ff}^{i}(t) = \tau_{ff}^{i-1}(t) + \alpha e^{i-1}(t), \tag{3}$$

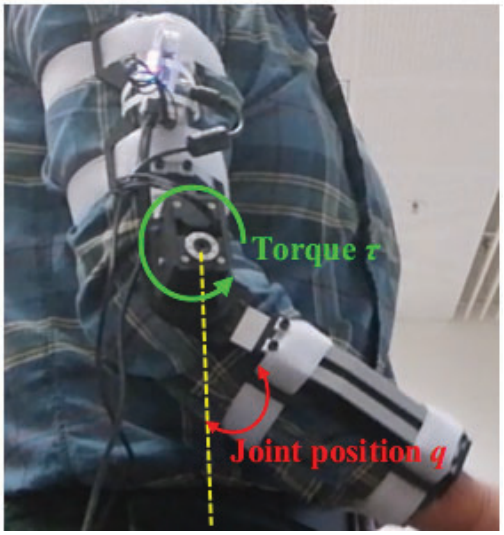


Fig. 1. Inverse dynamic model of an elbow exoskeleton (i.e., POW-EXO) and its wearer. The control torque of the POW-EXO is $\tau$ when its joint position is $q$ (see Eqs. (1) and (2)).

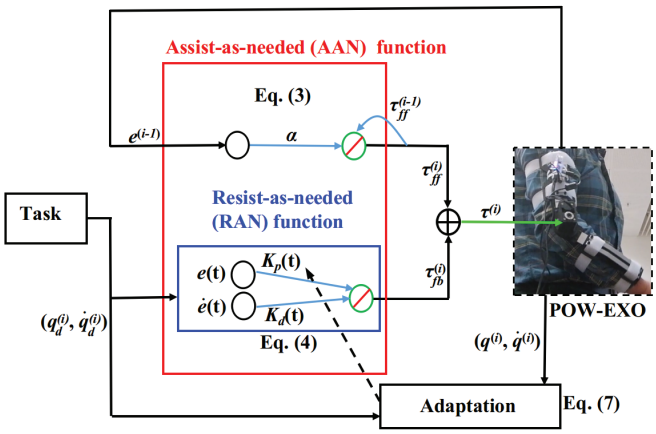


Fig. 2. A learning-based model for multifunctional elbow exoskeleton control. It consists of iterative learning (see Eq. (3)) and online impedance adaptation for predictive and variable compliant exoskeleton control. The dashed arrow line means that the impedance gains $K_p$ and $K_d$ are online adapted to involuntary and voluntary elbow movement (see Eqs. (4) and (7)).

where the feedforward term $\tau_{ff}$ at time $t$ of $i$-th iteration is determined by itself and position error $e$ at the previous iteration (see $e$ in Eq. (5)). The learning gain $\alpha$ is empirically set to $0.1$ for achieving fast stable control.

The principle of learning the feedforward term $\tau_{ff}$ is comparable to that used in human motor learning and control. Many human motor experiments have shown that humans learn to adapt to unknown dynamics in a predictive manner [21], [22]. Moreover, they adapt the mechanical impedance of their arms to position-based disturbances through sensory feedback (e.g., reflex) [23], [24]. Therefore, emulating such human motor learning requires the integration of predictive and adaptive learning control for unknown dynamics and different disturbances. Feedforward learning control can compensate for unknown dynamics, while control robustness against unexpected perturbations is enhanced by feedback-based online impedance adaptation.

### B. Online Impedance Adaptation Mechanism (OIAM)

The OIAM is designed to generate the feedback joint torque $\tau_{fb}$ at a trial. $\tau_{fb}$ is governed by a proportional-derivative (PD) rule [20],

$$\tau_{fb} = K_p(t)e(t) + K_d(t)\dot{e}(t), \tag{4}$$

where $K_p(t)$ and $K_d(t)$ denote the impedance gains, while $e(t)$ and $\dot{e}(t)$ represent the joint position and velocity errors given by,

$$e(t) = q(t) - q_d(t), \dot{e}(t) = \dot{q}(t) - \dot{q}_d(t), \varepsilon(t) = e(t) + \beta\dot{e}(t), \tag{5}$$

where $\varepsilon(t)$ denotes the joint tracking error with factor $\beta = 0.05$. In the OIAM, the impedance efforts and motion errors of the POW-EXO and its wearer are minimized over time period $T$ given by [20],

$$\begin{aligned} J_o(t) &= J_c(t) + J_p(t), \\ J_c(t) &= \frac{1}{2}\int_{t-T}^{t} (K_p(t))^2 + (K_d(t))^2, \\ J_p(t) &= \frac{1}{2}\int_{t-T}^{t} V(t), V(t) = I(\varepsilon(t))^2, \end{aligned} \tag{6}$$

This minimization leads to online impedance adaptation given by,

$$\begin{aligned} K_p(t) &= f(t)e(t), K_d(t) = f(t)\dot{e}(t), \\ f(t) &= \frac{\varepsilon(t)}{\beta(t)}, \beta(t) = \frac{a}{1 + b\varepsilon(t)^2}, \end{aligned} \tag{7}$$

where $\beta(t)$ is an adaptation scalar with the positive scalars $a = 0.2$ and $b = 5$. The values of $a$ and $b$ are chosen to adapt the response speed. All scalars, as well as the derivation of Eqs. (6) and (7) refer to our developed human-like impedance controller [25]. The stability proof of the online impedance adaptation law (i.e., Eq. (7)) can be seen in [25]. Note that the law has been validated in finger and elbow exoskeletons for exercise [20], [26], respectively.

### C. Multifunctional Exoskeleton Control

The implementation consists of assist- and resist-as-needed (AAN and RAN) control. Note that the AAN control is achieved by combining the ILM and OIAM, while only the OIAM is used for the RAN control (see Fig. 2). This is because only online impedance adaptation is required for the RAN control. The proposed control allows an elbow exoskeleton to provide the AAN and RAN to (stroke) patients in multi-stages [10], respectively. The AAN provides variable assistance to the patients with partial motor ability in repetitive exercises, in which their muscle controllability can be improved enough to initiate voluntary elbow movement. Further improvement will be facilitated by resist-as-needed exercises in late rehabilitation stage.

*1) Assist-as-needed (AAN) control:* In the AAN, the desired exoskeleton joint position $q_d$ is given by,

$$q_d(t) = -\frac{\pi}{2.0}|\sin(\frac{4\pi}{15.0}t)|[rad], t \in [0, 15.0][s], \tag{8}$$

where $15.0$ is the task time. Such desired position prevents elbow's over-extension (i.e., $q_d(t) > 0$), thereby guaranteeing wearer's safety.

In AAN control, parameter $\theta$ is learned to minimize the position error given by,

$$\begin{aligned} \text{minimize} \quad & |q_d - q(\theta)|, \theta = [\tau_{ff}, K_p, K_d]^T, \\ \text{subject to} \quad & I\ddot{q} + h(\dot{q}) + g(q) = \tau_s(\theta), \\ & \tau_s(\theta) = \tau(\theta) + \tau_h, \tau(\theta) = -(\tau_{ff} + \tau_{fb}), \end{aligned} \tag{9}$$

where $\tau_s$ denotes the sensed torque of the POW-EXO actuator. The iterative learning and online impedance adaptation mechanisms are integrated for learning predictive (i.e., $\tau_{ff}$) and adaptive (i.e., $(K_p, K_d)$) terms. Note that the AAN here enables the POW-EXO to generate motion assistance for compensating for involuntary (i.e., reflex) movement by various wearers. This is different from conventional AAN control that aims at maximizing wearers' involvement.

*2) Resist-as-needed (RAN) control:* Unlike the proposed AAN, only the adaptive terms $K_p$ and $K_d$ of the OIAM are online tuned for various exercisers. Therefore, the resistant torque is given by (rewritten from Eq. (2)),

$$\tau = -\tau_{fb} \quad . \tag{10}$$

This is because its desired exoskeleton joint position is fixed as,

$$q_d(t) = 0.0 \quad . \tag{11}$$

In contrast to the ANN, there is no cyclic tracking task. Moreover, the OIAM acts only as a virtual adjustable spring and damper for exercises. Greater resistant torque (i.e., $\tau$) results from larger joint position (i.e., $|e|$) and velocity (i.e., $|\dot{e}|$) errors (see Eq. (7)). In the RAN control, parameter $\Omega$ is online modulated to maximize wearer's exercise involvement quantified by,

$$\begin{aligned} \text{maximize} \quad & |\tau_s(\Omega)|, \Omega = [K_p, K_d]^T, \\ \text{subject to} \quad & I\ddot{q} + h(\dot{q}) + g(q) = \tau_s(\Omega), \\ & \tau_s(\Omega) = \tau(\Omega) + \tau_h, \tau(\Omega) = -\tau_{fb}(\Omega) \quad . \end{aligned} \tag{12}$$

## III. Experiments

The AAN and RAN control was validated on the POW-EXO worn by three healthy subjects[1] (i.e., S1-S3, see the experimental video link in [27]). The subjects were not physically supported by any ground-based framework, since the POW-EXO is a body-based exoskeleton [18].

### A. Assist-as-needed (AAN) Control

In the experiments, three subjects were asked to relax their muscles so as not to produce elbow joint torques. AAN torques were produced by the proposed model and POW-EXO, flexing and extending elbows to track a desired trajectory (see Eq. (8)). However, their involuntary (i.e., reflex) movements were not completely prevented, leading to the unknown interaction torque $\tau_h$ (see Eq. (1)) deviating from the trajectory. Moreover, such deviations were affected by uncertainties such as muscle fatigue and unrepeatable involuntary movement. Therefore, the proposed control aims at compensating for wearers' involuntary movement to minimize the position error (see Eq. (9)). This minimization is based on modulating the predictive (i.e., $\tau_{ff}$) and adaptive (i.e., $K_p$ and $K_d$) terms of the proposed model. The model was run for 16 trials on each subject.

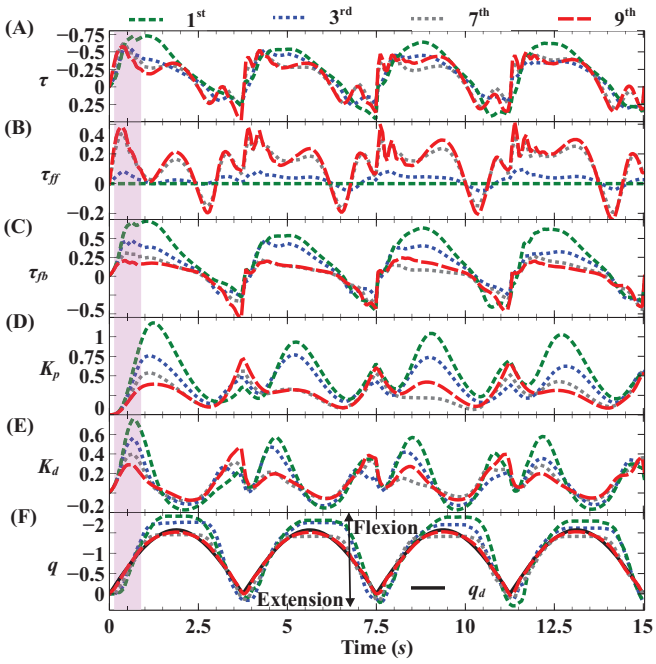


Fig. 3. Learning-based control for the POW-EXO in repetitive (i.e., elbow flexions and extensions) and AAN movements (see Eq. (8)). The control was run on the POW-EXO worn by subject S1. The results before learning are shown in the first and third trials, while those after learning are presented at the seventh and ninth trials. (A)-(C) Control $\tau$, feedforward $\tau_{ff}$, and feedback $\tau_{fb}$ torques. (D) and (E) Impedance parameters $K_p$ and $K_d$. (F) Real $q$ and desired $q_d$ joint positions (see Eq. (8)). The average joint position error $\overline{e}$ is indicated by the blue line in row 1 of Fig. 4.

[1]The subjects consent to the participation in the experiments, which have been approved by the Research Ethics Committee of the University of Southern Denmark (case no.: 20/70421).

The learning-based control starts at the feedback control without feedforward control (i.e., $\tau_{ff}^1 = 0.0$, see Fig. 3 (B)) in the first trial. We can see that the feedforward torque $\tau_{ff}$ increases, while the feedback $\tau_{fb}$ decreases over the learning trials (e.g., see the magenta area in Fig. 3). The decreased feedback $\tau_{fb}$ results from a reduction in task error $e$. The increased feedforward and decreased feedback contributions by the proposed model guarantee the online control predictivity and adaptivity of the POW-EXO. As a result, the co-contributions reduce the average position error $\overline{e}$ over trials (see the blue line in row 1 of Fig. 4). The average joint position error $\overline{e}$ is given by [9], [13],

$$\overline{e}^i = \frac{\sum_{t=1}^{t=m} |q^i(t) - q_d^i(t)|}{m} [rad], \tag{13}$$

where $i$ is the trial ID, and $m$ is the time step. A small $\overline{e}$ indicates good AAN control.

The proposed control enables the POW-EXO to online learn the predictive (i.e., $\tau_{ff}$) and adaptive (i.e., $K_p$ and $K_d$) terms in the AAN tasks (see rows 2-4 of Fig. 4). As a result, it leads to variable predictive and compliant motion adaptation for various wearers. We can see that the medians of $\tau_{ff}$, $K_p$, and $K_d$ increase from subjects S1 to S3. Moreover, the deviation of the predictive (i.e., $\tau_{ff}$) term is larger than those of the adaptive (i.e., $K_p$ and $K_d$). This is because the learning-based control starts with online impedance adaptations. By contrast, its predictive learning is initialized from scratch (i.e., $\tau_{ff}^1 = 0.0$). In the proposed model, iterative learning and online impedance adaptation mechanisms compensate for unknown dynamics and interaction of human-exoskeleton shared control, respectively. Note that the proposed learning-based control focuses on AAN adaptations for various wearers, rather than traditional AAN control for maximizing wearer's involvement [13].

### B. Resist-as-needed (RAN) Control

Unlike the AAN control, the RAN aims at maximizing wearers' involvement (i.e., $|\tau_s|$, see Eq. (12)) only using the online impedance adaptation mechanism (OIAM) (see Section II-B). In the experiments, three subjects were asked to flex their elbows as much as possible. In elbow flexions, the resistant torque $\tau$ increases with rising flexion angles and velocities, resulting in large impedance terms $K_p$ and $K_d$ (see Eq. (7)). Therefore, $K_p$ and $K_d$ of the OIAM are online adapted to generate variable resistance against different wearers. The OIAM acts as a virtual adjustable spring and damper (see Eq. (4)).

For instance, the OIAM enables the POW-EXO to produce variable resistant torque $\tau$ against subjects S2-3 via online modulating the impedance terms $K_p$ and $K_d$ (see Fig. 5 (A)-(C)). The resistance increases with the rising impedance terms (see the green area in Fig. 5). The online resistance adaptation control is smooth, because its impedance terms are online modulated based on a stable control law (see Eq. (7)). According to the law, the impedance terms $K_p$ and $K_d$ increase with the increments of $|e|$ and $|\dot{e}|$, respectively.

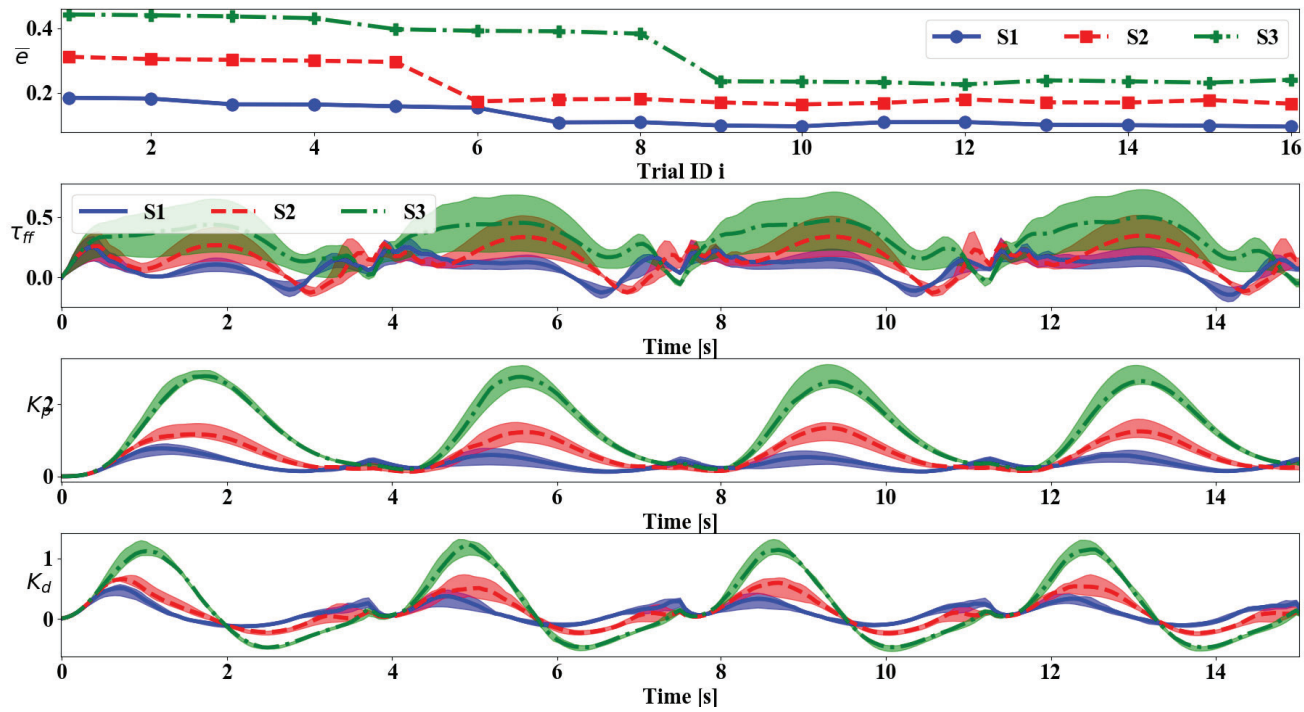


Fig. 4. Statistical results of the proposed model for AAN tasks. In the tasks, three subjects (i.e., S1-S3) wore the POW-EXO, and 16 trials were run for each subject. Rows 1-4 illustrate the average position error $\overline{e}$ $[rad]$, the medians and deviations of learned feedforward torque $\tau_{ff}$, and impedance parameters $K_p$ and $K_d$. Blue, red, and green colors depict the results of subjects S1-S3, respectively.

Interestingly, the prolonged repetitions with large position and velocity deviations result in strong resistances (see S3 in Fig. 5). These resistances are led by the rising impedance gains $K_p$ and $K_d$ in terms of their online adaptation law (see Eq. (7)).

The wearers' involvement was maximized through the OIAM. The subject's exercise involvement is quantified by [10], [28],

$$\overline{\tau}_s^i = \frac{\sum_{t=1}^{t=m} |\tau_s(t)|}{m}, \tag{14}$$

where $i$ is the trial ID, and $m$ is the time step. A deep involvement is indicated by a strong torque feedback (i.e., $\tau_s$). We can see that subject S1 obtained deeper involvement, compared to other subjects (see row 1 of Fig. 6). Such involvement results from larger impedance terms (i.e., $K_p$ and $K_d$, see rows 2-3 of Fig. 6). The resulting impedance terms and resistant torque $\tau$ increase with rising position and velocity errors (see rows 4-5 of Fig. 6), leading to strong torque feedback (i.e., $\tau_s$). Therefore, the OAIM makes the POE-EXO generate resist-as-needed elbow motions against various wearers in exercises (see Fig. 7).

Interestingly, the proposed RAN control allows the exoskeleton to quantify the mechanical impedance of interactive elbow control. This quantification indicates that interactive impedance of elbows may vary over time (see $K_p(t)$ and $K_d(t)$ in Fig. 6), but eventually their averages converge at specific ranges (see $\overline{K_p}(t)$ and $\overline{K_d}(t)$ in Figs. 8 and 9).

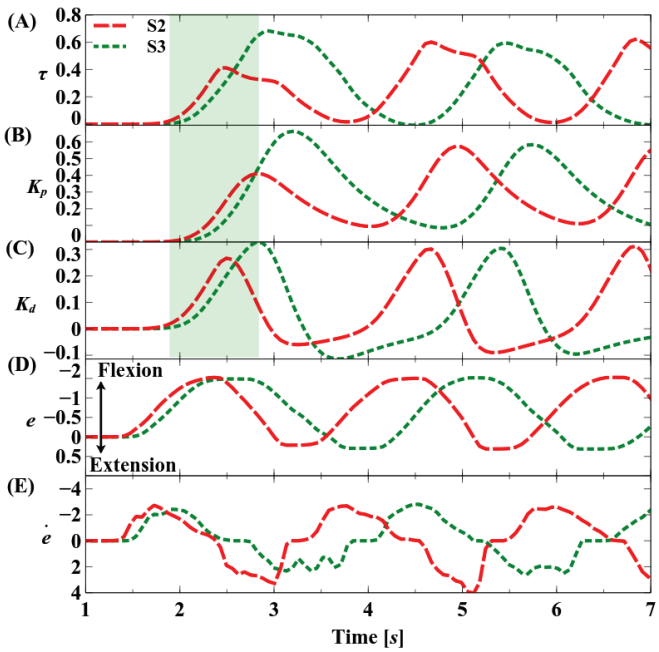


Fig. 5. Variable resistance against subjects S2-3 in exercises (i.e., elbow flexions and extensions). (A) Control $\tau$. (B) and (C) Impedance parameters $K_p$ and $K_d$. (D) and (E) Position $e$ and velocity $\dot{e}$ errors. The green area illustrates the resistant torque $\tau$ increasing with the impedance terms $K_p$ and $K_d$.

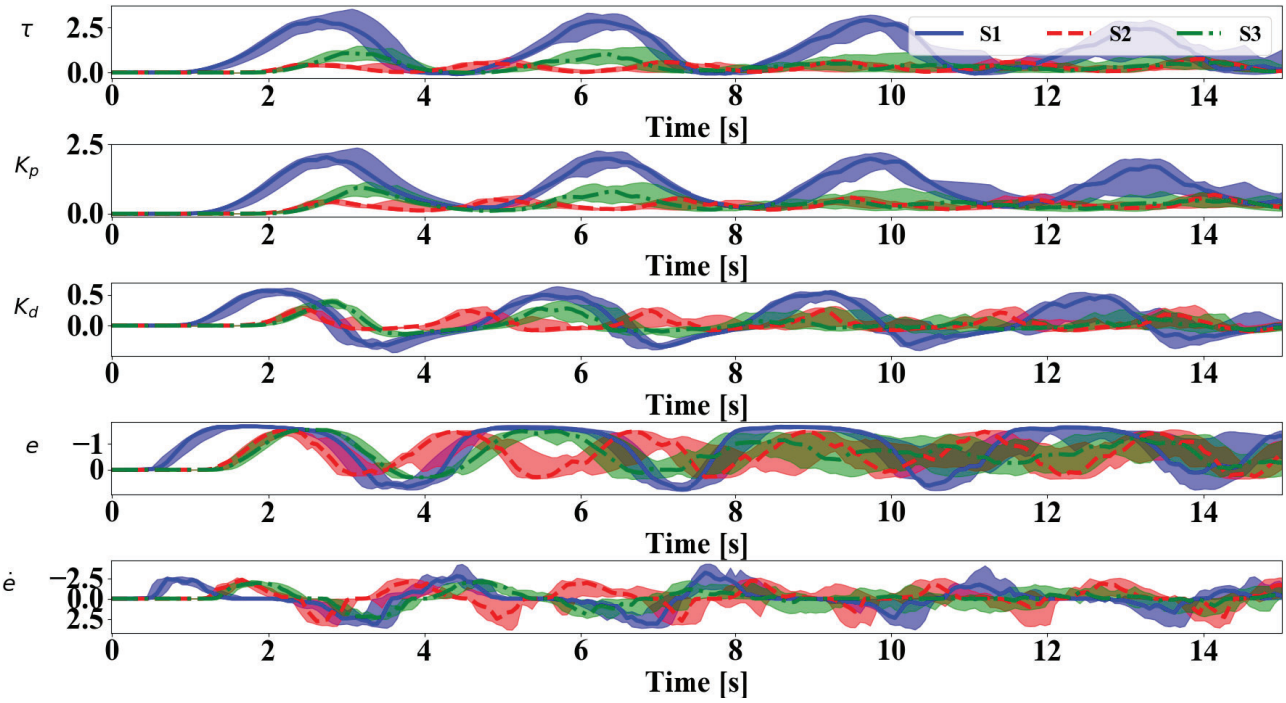


Fig. 6. Statistical results of the proposed online impedance adaptation control for RAN tasks. In the tasks, three subjects (i.e., S1-S3) wore the POW-EXO, and 16 trials were run for each subject. Rows 1-5 illustrate the medians and deviations of the joint control torque $\tau$ (see Eq. 10), impedance parameters $K_p$ and $K_d$, the position $e$ and velocity $\dot{e}$ errors (see Eq. (5)). Blue, red, and green colors depict the results of subjects S1-S3, respectively.

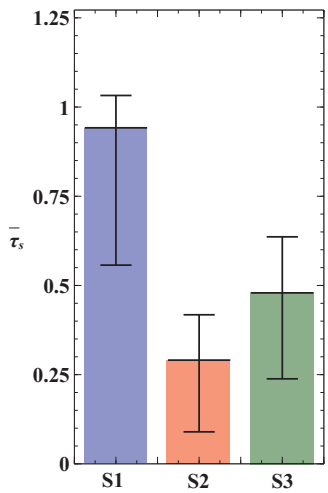


Fig. 7. Exercise involvement (see Eq. (14)) of subjects S1-S3.

The indication may reveal interactive mechanical impedance variance and invariance in human motor control [23], [24]. Here the average mechanical impedance $\overline{K_p}(t)$ and $\overline{K_d}(t)$ are given by,

$$\overline{K_{p,d}}(t) = \frac{\overline{K_{p,d}}(t-\Delta t)(n-1) + K_{p,d}(t)}{n}, \Delta t \approx 0.06(s), \tag{15}$$

where $\overline{K_p}(t)$ and $\overline{K_d}(t)$ are the moving averages of mechanical impedance parameters $K_p(t)$ and $K_d(t)$ [24]. $\Delta t$ is the time step, while $n$ is the step length. The same controller has been used to quantify the finger muscle fatigue [29].

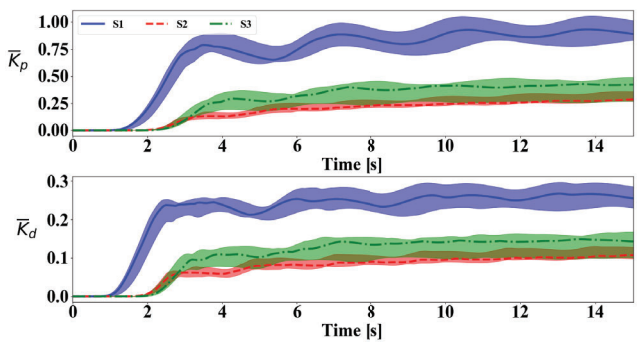


Fig. 8. Average mechanical impedance of interactive elbow control. Rows 1-2 show the average impedance $\overline{K}_p$ and $\overline{K}_d$ (see Eq. 15) of the RAN controller with respect to subjects S1-3.

The presented OIAM can free humans from manually tuning mechanical impedance of exoskeletons for various subjects. Moreover, the control outperforms large impedance controllers in terms of its friendly interaction control and actuator dysfunction prevention. The comparison can be seen in [20].

### C. Control Comparison

The proposed learning-based control model (i.e., LBCM) outperforms conventional trial-error and iterative impedance

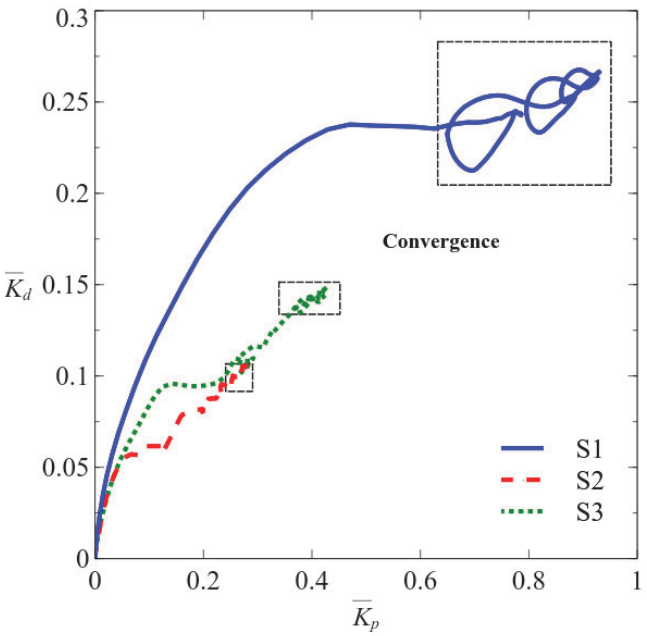


Fig. 9. Variance and invariance (convergence, see the black dashed rectangles) of average mechanical impedance of interactive elbow control with respect to subjects S1-3. The blue, red, green lines denote the medians of the average impedance $\overline{K}_p$ and $\overline{K}_d$ (see Fig. 8).

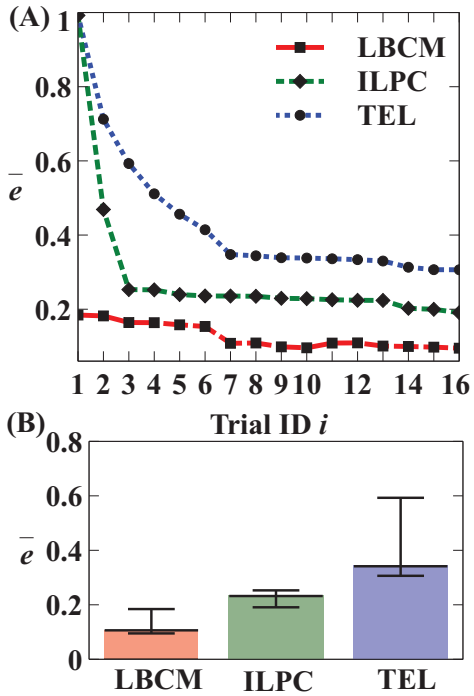


Fig. 10. Control comparison between our and state-of-art controllers. It was performed on subject S1.

learning (i.e., TEL and ILPC) in the AAN task (see section III-A). The comparison was performed on subject S1. We can see that the LBCM enables the POW-EXO to achieve fewer position errors (see $\overline{e}$ in Fig. 10 (A) and (B)), compared to the ILPC and TEL. This is because its impedance adaptation mechanism online modulates interaction impedance terms based on position and velocity errors. Such a modulation makes the LBCM start at a smaller position error, compared to the ILPC and TEL. Note that the TEL and ILPC were revised based on their origins, such that the predictive (i.e., $\tau_{ff}$) and adaptive (i.e., $K_p$ and $K_d$) terms are learned for a suitable comparison (see their implementations in Section V).

## IV. Discussion and Conclusion

The majority of this paper focuses on our novel learning-based model achieved for multifunctional elbow exoskeleton control. However, it is helpful to discuss and conclude its contribution to state-of-art (SOA) methods and future extensions. First, most SOA methods rely heavily on additional external sensing such as EMG and force [14], [15]. However, these sensors are subject to the following intrinsic limitations: 1) Their signals are inherently noisy and vary in accordance with different applications [12]; 2) An activated muscle contributes to various limb movements, leading to the inaccurate prediction of wearers' intentions [30]; 3) The external sensing becomes less precise when wearers perspire and sensors fall out [5]; 4) Estimating the interaction torques from the EMG and force signals of patients can be difficult owing to an abnormal EMG-torque relationship [31], [32]. These limitations may reduce learning and control robustness of state-of-art position/velocity [19], [33] and torque [9], [17] methods for different tasks and wearers. To address these limitations, predictive learning and online (impedance) adaptation control were integrated to exclude EMG and force sensors in our proposed model. Its implementation relied only on the joint position and velocity feedback, thereby enhancing exoskeleton learning-based control robustness (see detailed comparisons in TABLE II). Our impedance control is better suited for friendly exoskeleton and human interaction control [10], [11], compared to position/velocity (admittance) control [19], [33]. Second, our proposed model allows an elbow exoskeleton to provide two functions to patients in different rehabilitation stages. This greatly increases exoskeletons' affordability and productivity in home-based applications, compared to conventional single-function one.

Our core scientific contribution is online stiffness and damping adaptation mechanism, providing a novel way to achieve multifunctional exoskeleton control, as well as interactive (mechanical) impedance quantification (IIQ, see TABLE II). The mechanism enables our proposed method to 1) have lower sensory dependence than classical angle/velocity (admittance) control [19], [33]; 2) handle non-repeatable arm movement dynamics better than conventional variable impedance controllers [9], [13] (see Fig. 10). Their stiffness $K_p$ and damping $K_d$ gains are modulated based on previous (trial) tracking errors (see Eqs. (16) and (17)). 'Non-repeatable' here means a subject would not repeat the same involuntary arm feedback in task trials, leading to unknown and different arm dynamics.

Despite the presented effectiveness, the proposed model still requires improvement. First, the model can be extended for multi-joint exoskeleton control. A possible solution is to extend the model for Cartesian control of multi-joint exoskele-

TABLE II
COMPARISON BETWEEN OUR PROPOSED AND STATE-OF-ART METHODS

| Method | | Ours | Lotti [19] | Zhijun [33] | Li [9] | Proietti [13] |
|---|---|---|---|---|---|---|
| Function | AAN | Yes | Yes | Yes | Yes | Yes |
| | RAN | Yes | No | No | No | No |
| | IIQ | Yes | No | No | No | No |
| Control | Torque | Yes | No | No | Yes | Yes |
| | Angle/velocity | No | Yes | Yes | No | No |
| | Predictive feed-forward learning | Yes | No | No | Yes | Yes |
| | Online stiffness and damping adaptation | Yes | No | No | No | No |
| Required sensing | EMG | No | Yes | No | No | No |
| | Force/torque | No | Yes | Yes | No | No |
| | Joint position | Yes | Yes | Yes | Yes | Yes |
| | Joint velocity | Yes | Yes | Yes | Yes | Yes |

tons. Second, the model can be revised for applications to hybrid exoskeletons consisting of passive and active actuation. The terms (e.g., impedance $K_p$ and $K_d$) for active actuation will be online learned to work with passive (e.g., physical springs) in terms of energy-efficient exoskeleton control.

## V. APPENDIX: COMPARED LEARNING-BASED CONTROL

### A. Trial-and-error Learning (TEL)

Comparable to human sensorimotor learning control, motor errors are used to tune motor commands in the TEL. The trial error is used to increase exoskeleton assistance, if a subject does not achieve task minimization. Based on the error $\overline{e}^{i-1}$, parameters $\theta = [\tau_{ff}, K_p, K_d]^T$ at the $i$-th trial are learned by:

$$\begin{aligned} &\tau_{ff}^i = \tau_{ff}^{i-1} + \lambda \overline{e}^{i-1}, K_p^i = K_p^{i-1} + \lambda \overline{e}^{i-1}, \lambda = 0.2, \\ &K_d^i = \mu k_p^i, \mu = 0.2, \end{aligned} \tag{16}$$

where the learning rate $\lambda$ and ratio $\mu$ were empirically chosen to achieve fast stable learning-based exoskeleton control. The learning law is revised from the one presented in [13], in which there is no feedforward term (i.e., $\tau_{ff}$). Although the original law was proposed to maximize subjects' involvement, it is still a classical trial-and-error learning impedance control model for adaptive control aimed by the presented AAN control task.

### B. Iterative Learning Impedance Control (ILPC)

In contrast to TEL learning (see Eq. (16)), parameters $\theta = [\tau_{ff}, K_p, K_d]^T$ in the ILPC are tuned over the time steps of a trial. The ILPC mimics human sensorimotor learning in physical interaction control [24]. Humans tend to modulate their feedforward motor and feedback impedance control commands in unstable interactive tasks (e.g., tooling) [23]. The modulation is based on the simultaneous minimization of feedforward and feedback effort, leading to an iterative learning impedance control law,

$$\begin{aligned} &\tau_{ff}^i(t) = \tau_{ff}^{i-1}(t) + Q_F[\varepsilon(t) - \beta(t)\tau_{ff}^{i-1}(t)], \\ &K_p^i(t) = K_p^{i-1}(t) + Q_P[\varepsilon(t)e(t) - \beta(t)K_p^{i-1}(t)], \\ &K_d^i = K_d^{i-1}(t) + Q_D[\varepsilon(t)\dot{e}(t) - \beta(t)K_d^{i-1}(t)], \end{aligned} \tag{17}$$

where learning rates are set as: $Q_F = 0.5$, $Q_P = 0.2$, and $Q_D = 0.2$. The rates were empirically chosen to achieve fast stable elbow exoskeleton control. The stability proof of the learning law (i.e., Eq. (17)) refers to [34]. A similar learning law has been applied to robotic rehabilitation [9]. The position $e(t)$ and velocity $\dot{e}(t)$ errors can be seen in Eq. (5), while factor $\beta(t)$ is given in Eq. (7).

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